Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Simplification of algebraic expressions: order of operations, brackets and word problems. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Second Term
Week	9
Topic	Simplification of Algebraic Terms
Subtopic	Simplification of algebraic expressions: order of operations, brackets and word problems
Theme	—
Curriculum objectives	• carry out order of operation in algebra • insert/remove bracket and simplify expression • solve word problems involving algebraic expression
Previous knowledge	• Addition and subtraction of algebraic terms • Multiplication and division of algebraic terms
Lesson sequence	• carry out order of operation in algebra — the teacher solves some questions in order of operation on the board; the student listen, contribute, and where necessary ask questions where they do not understand; the student copy worked examples into their notes; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, mark their notes and mark correction. Set for the students: 1. $4n + 3n \times 10$ 2. $32 + 8x \div 2$ 3. $3 \times 5x + 4x \div 2$ 4. $36y \div 6 + y + 2 \times 4y$ • insert/remove bracket and simplify expression — the teacher solves some questions on insertion, removal of bracket on the board; the student listen, ask questions on their areas of difficulty; the student copy worked examples into their notes; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, marks their notes and makes corrections. Set for the students: (a) $(12a - 7) + (4 - 5a)$ (b) $(8x - 2y) - (9x - 4y)$ • solve word problems involving algebraic expression — the teacher solves some questions on word problems involving algebraic expression on the board; the student listen, ask questions where they do not understand; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, mark their notes and mark correction. Set for the students: 1. A pen cost h naira and a crayon cost ₦20 less than a pen. (a) Find the cost of a crayon (b) Find the cost of a crayon and a pen 2. Envelopes cost d naira per pack, 4 note pad cost 20 naira more than a pack of envelopes. Find the total cost of a pack of envelope and notepad.
Instructional materials	• Chart showing algebraic expression.
Evaluation	• Simplify (a) $14pq \times r$ (b) $12x^2y \div 3x$ • Simplify $7 \times 3a - (3a + 5a) \times 2$

	• Simplify $12-(7-2f)-(3-f)$ • Simplify $(8a-5)+(14-3a)$ • A pen and a pencil altogether cost x naira. If a pen cost y naira, find the cost of 3 pencils.

Lesson objectives

1.	carry out order of operation in algebra
2.	insert/remove bracket and simplify expression
3.	solve word problems involving algebraic expression

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Simplification of algebraic expressions: order of operations, brackets and word problems

Class: JSS1 **Subject:** Mathematics **Week:** 9

Section 1: Getting the Order Right in Algebra

When an algebraic expression mixes different operations, you cannot just work from left to right — the order you do them in changes the answer. In algebra, just as in arithmetic, you follow a fixed order:

1. **B**rackets first — work out whatever is inside them.
2. **O**rders (powers and roots), if any.
3. **D**ivision and **M**ultiplication, from left to right.
4. **A**ddition and **S**ubtraction, from left to right.

You can remember this as **BODMAS**. The key rule for algebra is that multiplication and division are done *before* addition and subtraction. So in an expression like $4n + 3n \times 10$, you do not add $4n + 3n$ first — you must multiply $3n \times 10$ before you add anything. Brackets, when they appear, override everything and are worked out first.

Simplify $4n + 3n \times 10$.

There are no brackets, so multiplication comes first.

- Multiply first: $3n \times 10 = 30n$
- Now add: $4n + 30n = 34n$

So $4n + 3n \times 10 = 34n$. If you had wrongly added first you would have got $7n \times 10 = 70n$ — a very different result, so the order really matters.

Simplify $7 \times 3a - (3a + 5a) \times 2$.

Here there is a bracket, and BODMAS says it comes first.

- Work out the bracket: $3a + 5a = 8a$
- The expression is now $7 \times 3a - 8a \times 2$
- Multiply first: $7 \times 3a = 21a$ and $8a \times 2 = 16a$
- Now subtract: $21a - 16a = 5a$

So $7 \times 3a - (3a + 5a) \times 2 = 5a$.

Practice

1. Simplify $4n + 3n \times 10$.
2. Simplify $32 + 8x \div 2$.
3. Simplify $3 \times 5x + 4x \div 2$.
4. Simplify $36y \div 6 + y + 2 \times 4y$.
5. Simplify $7 \times 3a - (3a + 5a) \times 2$.

Section 2: Removing and Inserting Brackets in Algebra

Brackets are used in algebra to group terms together. When a whole expression sits inside a bracket, everything inside must be treated as one unit. To simplify an expression containing brackets, you first remove the brackets, and then you add or subtract the like terms.

The sign that sits just in front of a bracket tells you what to do with the terms inside:

- When a plus sign sits in front of the bracket, you may remove the bracket and write each term inside with the sign it already has.
- When a minus sign sits in front of the bracket, removing the bracket changes the sign of every term inside: a plus becomes a minus, and a minus becomes a plus.

Think of it this way: subtracting a group is the same as subtracting each member of that group one by one. So $-(a+b)$ means $-a-b$, and $-(a-b)$ means $-a+b$.

Once the brackets are gone, you simplify by collecting like terms — terms that have exactly the same letter. Add or subtract their numbers and keep the letter unchanged.

Example 1: Simplify $(12a-7)+(4-5a)$.

A plus sign is in front of each bracket, so we remove the brackets and keep every sign the same:

$$(12a-7)+(4-5a)=12a-7+4-5a$$

Now collect like terms. The terms in a are $12a$ and $-5a$, and the plain numbers are -7 and 4:

$$12a-5a-7+4=7a-3$$

So $(12a-7)+(4-5a)=7a-3$.

Example 2: Simplify $(8x-2y)-(9x-4y)$.

The first bracket has a plus sign in front, so its terms stay as they are. The second bracket has a minus sign in front, so every sign inside it must change: $9x$ becomes $-9x$ and $-4y$ becomes $+4y$.

$$(8x-2y)-(9x-4y)=8x-2y-9x+4y$$

Collect the like terms:

$$8x-9x-2y+4y=-x+2y$$

So $(8x-2y)-(9x-4y)=-x+2y$, which is often written $2y-x$.

Inserting a bracket is the reverse process. When you need to put brackets around part of an expression, you work out what sign must go in front of the bracket so that the expression inside changes correctly.

- To insert a bracket with a plus sign in front, simply write the bracket and keep every term's sign the same as it was.
- To insert a bracket with a minus sign in front, every term that goes inside must have its sign changed — because removing that bracket later would flip the signs back to what you started with.

For example, suppose you want to factor a pair of terms out of $10a-5b-3$ so that the last two terms sit in a bracket after a minus sign. You want to write

$$10a-(5b+3)$$

Check it: removing the bracket gives $10a-5b-3$, exactly the original expression. Because the minus sign is in front, the $5b$ and the 3 both had their signs changed (both were plus in the bracket, and both come out as minus). That is how you know the bracket was inserted correctly.

Practice

1. Simplify $12-(7-2f)-(3-f)$.
2. Simplify $(8a-5)+(14-3a)$.
3. Insert a bracket around the last two terms of $14a-6b-4$ so that a minus sign stands in front of the bracket, keeping the value the same.

Section 3: Turning Word Problems into Algebraic Expressions

A word problem gives you information in sentences, and your job is to turn those sentences into algebra before you do any simplifying. The key is to read carefully and decide what each part means.

Here is how to translate common phrases:

- "A pen costs h naira" — the unknown quantity h stands for the pen's price.
- "20 less than a pen" — take the pen's price and subtract 20, so it becomes h − 20.
- "A crayon costs ₦20 less than a pen" — the crayon costs h − 20 naira.
- "The total cost of the two" — add the two costs together.
- "3 more than something" — add 3; "3 less than something" — subtract 3.
- "twice a number" — multiply by 2; "half of a number" — divide by 2.

So a sentence like "a pencil costs ₦5 less than a pen that costs h naira" becomes h − 5. Once you have written the algebra, you simplify it the same way you have been doing all along.

Example 1
A pen costs h naira and a crayon costs ₦20 less than a pen.

(a) Find the cost of a crayon.
The crayon costs ₦20 less than the pen, so subtract 20 from the pen's price:
Cost of a crayon = h − 20 naira.

(b) Find the total cost of a crayon and a pen.
Add the pen's cost and the crayon's cost:
Total = h + (h − 20) = h + h − 20 = 2h − 20 naira.

Example 2
Envelopes cost d naira per pack, and 4 notebooks cost ₦20 more than a pack of envelopes. Find the total cost of a pack of envelopes and the notebooks.

The envelopes cost d naira.
The notebooks cost ₦20 more than the envelopes, so they cost d + 20 naira.
Total cost = d + (d + 20) = d + d + 20 = 2d + 20 naira.

Notice in both examples that you write the phrase as an expression first, then collect the like terms to simplify.

Practice

1. A pencil costs ₦x and a pen costs ₦8 more than a pencil. Write an expression for the total cost of the pencil and the pen, and simplify it.
2. A packet of biscuits costs k naira and a loaf of bread costs ₦12 less than a packet of biscuits. Find, in terms of k, the total cost of the biscuit packet and the loaf of bread.
3. A pen and a pencil together cost ₦x. If the pen costs ₦y, write an expression for the cost of 3 pencils.

Check your work

Attempt every question before you read this page.

Getting the Order Right in Algebra

1. 34n

   Multiplication comes before addition: 3n × 10 = 30n, then 4n + 30n = 34n.
2. 32 + 4x

   Division comes before addition: 8x ÷ 2 = 4x, then 32 + 4x (the 32 and the 4x are unlike terms, so they cannot be combined).
3. 17x

   Do the multiplication and division first, left to right: 3 × 5x = 15x and 4x ÷ 2 = 2x. Then add: 15x + 2x = 17x.
4. 15y

   Do the division and multiplication first, left to right: 36y ÷ 6 = 6y and 2 × 4y = 8y. The expression becomes 6y + y + 8y. Now add: 6y + y + 8y = 15y.
5. 5a

   Work out the bracket first: 3a + 5a = 8a. Now the expression is 7 × 3a − 8a × 2. Multiply first: 7 × 3a = 21a and 8a × 2 = 16a. Then subtract: 21a − 16a = 5a.

Removing and Inserting Brackets in Algebra

1. $2+3f$

   The first number 12 has no bracket around it, so it stays. The first bracket has a minus sign in front, so $7-2f$ becomes $-7+2f$. The second bracket also has a minus sign in front, so $3-f$ becomes $-3+f$. Together:

   $$12-(7-2f)-(3-f)=12-7+2f-3+f$$

   Collect the plain numbers $12-7-3=2$, and collect the f terms $2f+f=3f$:

   $$12-7+2f-3+f=2+3f$$

2. $5a + 9$

   A plus sign is in front of each bracket, so remove them and keep every sign the same:

   $$(8a - 5) + (14 - 3a) = 8a - 5 + 14 - 3a$$

   Collect the a terms $8a - 3a = 5a$, and collect the numbers $-5 + 14 = 9$:

   $$8a - 5 + 14 - 3a = 5a + 9$$

3. $14a - (6b + 4)$

   Write the first term $14a$ as it is, then place the bracket after the minus sign. The two terms that go inside are $-6b$ and -4, but because a minus sign will sit in front of the bracket, their signs must be changed inside the bracket: $-6b$ becomes $+6b$ and -4 becomes $+4$. So the result is

   $$14a - (6b + 4)$$

   Check by removing the bracket: $14a - 6b - 4$, which matches the original expression.

Turning Word Problems into Algebraic Expressions

1. x + (x + 8) = 2x + 8 naira

   The pen costs ₦8 more than the pencil, so the pen costs x + 8. The total is the pencil plus the pen: x + (x + 8) = x + x + 8 = 2x + 8.

2. k + (k − 12) = 2k − 12 naira

   The bread costs ₦12 less than the biscuits, so it costs k − 12. The total is k + (k − 12) = k + k − 12 = 2k − 12.

3. 3(x − y) naira

   The pen and pencil together cost x, and the pen costs y, so one pencil costs x − y. Three pencils cost 3 × (x − y) = 3(x − y) naira.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-1ab85995	**Date**	
Subtopic	Simplification of algebraic expressions: order of operations, brackets and word problems	**Rater**	

The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____